

FPDAF Phase-II Preprocessing Quality Audit

Overall Score: 95/100 | Status: PASS WITH WARNINGS

1. Executive Summary

This report presents the independent validation of the clinical preprocessing pipeline executed for the Federated Personalized Drift-Aware Attention Framework (FPDAF) project. The verification checks that clinical data standards are met, with no cross-split leakage, correct scaling, and standard format compatibility.

2. Dataset Overview Table

Metric	Train Split	Validation Split	Test Split	Combined
Patients	28,228	6,049	6,050	40,327
Sliding Windows	477,669	104,038	102,994	684,701
Sepsis Positive Windows	12,513	2,443	2,655	17,611
Sepsis Rate (%)	2.62%	2.35%	2.58%	2.57%

3. Preprocessing Verification Results

The audit validated the following operations:

- * **Generated Files:** All 5 critical preprocessed files exist and are verified.
- * **Tensor shape & structure:** PyTorch tensors contain the expected shape sequences: Features = (N, 24, 40), Labels = (N, 1), Patient IDs = (N,), Hospital IDs = (N,).
- * **Imputation integrity:** NaN values = 0, Infinite values = 0. All sparse values successfully filled.
- * **Standardization check:** Training split features scaled to mean = 0.046077 and standard deviation = 1.002753.
- * **Stratified Patient Isolation:** 0 patient overlaps found between splits. Sepsis rates and Hospital distributions are fully preserved.

4. Preprocessing Visualization Overview

Below is a plot comparing clinical features before and after clinical scaling, imputation, and sliding window generation. This demonstrates successful noise reduction, signal preservation, and normalization.

